



Service Bulletin Number: 5676975

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Cylinder Liner Reuse Guidelines

## Cylinder Liner Reuse Guidelines

### Introduction

This document provides reuse guidelines for cylinder liners and supplements the information provided in the Service Manuals listed in Table 1 below. These guidelines are to be consulted when determining if cylinder liners are acceptable for reuse.

### Product Affected

- ISL G CM2180
- ISX12 G CM2180 EJ
- ISX12N CM2380 X12B
- L9N CM2380 L124B/L9N CM2380 L130B
- L9N CM2380 L142B

### Publications Affected

Table 1, Publications Affected

| Manual Type | Engine | Bulletin Number | Procedure Title | Procedure Number | Procedure Section |
|-------------|--------|-----------------|-----------------|------------------|-------------------|
|-------------|--------|-----------------|-----------------|------------------|-------------------|

|                   |                          |         |                                      |                                                                                           |    |
|-------------------|--------------------------|---------|--------------------------------------|-------------------------------------------------------------------------------------------|----|
| Service<br>Manual | ISX12N<br>CM2380<br>X12B | 5504152 | Cylinder<br>Block                    | Refer to<br>Procedure<br>001-026<br>(/qs3/pubsys2/xml/en/procedures/525/525-001-026.html) | 1  |
|                   |                          |         | Cylinder<br>Block and<br>Liner Seats | Refer to<br>Procedure<br>001-027<br>(/qs3/pubsys2/xml/en/procedures/525/525-001-027.html) |    |
|                   |                          |         | Cylinder<br>Liner                    | Refer to<br>Procedure<br>001-028<br>(/qs3/pubsys2/xml/en/procedures/525/525-001-028.html) |    |
|                   |                          |         | Engine<br>Compression                | Refer to<br>Procedure<br>014-012<br>(/qs3/pubsys2/xml/en/procedures/525/525-014-012.html) | 14 |

|                   |                      |         |                                      |                                                                                                              |    |
|-------------------|----------------------|---------|--------------------------------------|--------------------------------------------------------------------------------------------------------------|----|
| Service<br>Manual | ISX12 G<br>CM2180 EJ | 4310682 | Cylinder<br>Block                    | Refer to<br>Procedure<br>001-026<br>(/qs3/pubsy<br>s2/xml/en/pr<br>ocedures/18<br>1/181-001-<br>026.html)    | 1  |
|                   |                      |         | Cylinder<br>Block and<br>Liner Seats | Refer to<br>Procedure<br>001-027<br>(/qs3/pubsy<br>s2/xml/en/pr<br>ocedures/18<br>1/181-001-<br>027.html)    |    |
|                   |                      |         | Cylinder<br>Liner                    | Refer to<br>Procedure<br>001-028<br>(/qs3/pubsy<br>s2/xml/en/pr<br>ocedures/18<br>1/181-001-<br>028-tr.html) |    |
|                   |                      |         | Engine<br>Compression                | Refer to<br>Procedure<br>014-012<br>(/qs3/pubsy<br>s2/xml/en/pr<br>ocedures/18<br>1/181-014-<br>012.html)    | 14 |

|                   |                 |         |                                      |                                                                                                 |    |
|-------------------|-----------------|---------|--------------------------------------|-------------------------------------------------------------------------------------------------|----|
| Service<br>Manual | ISL G<br>CM2180 | 4021649 | Cylinder<br>Block                    | Refer to<br>Procedure<br>001-026<br>(/qs3/pubsys2/xml/en/procedures/45/45-001-026.html)         | 1  |
|                   |                 |         | Cylinder<br>Block and<br>Liner Seats | Refer to<br>Procedure<br>001-027<br>(/qs3/pubsys2/xml/en/procedures/45/45-001-027.html)         |    |
|                   |                 |         | Cylinder<br>Liner                    | Refer to<br>Procedure<br>001-028<br>(/qs3/pubsys2/xml/en/procedures/45/45-001-028.html)         |    |
|                   |                 |         | Engine<br>Compression                | Refer to<br>Procedure<br>014-012<br>(/qs3/pubsys2/xml/en/procedures/45/45-014-012-tr-islg.html) | 14 |

|                   |                                            |         |                                      |                                                                                                           |    |
|-------------------|--------------------------------------------|---------|--------------------------------------|-----------------------------------------------------------------------------------------------------------|----|
| Service<br>Manual | L9N CM2380<br>L124B/L9N<br>CM2380<br>L130B | 5467492 | Cylinder<br>Block                    | Refer to<br>Procedure<br>001-026<br>(/qs3/pubsy<br>s2/xml/en/pr<br>ocedures/52<br>1/521-001-<br>026.html) | 1  |
|                   |                                            |         | Cylinder<br>Block and<br>Liner Seats | Refer to<br>Procedure<br>001-027<br>(/qs3/pubsy<br>s2/xml/en/pr<br>ocedures/52<br>1/521-001-<br>027.html) |    |
|                   |                                            |         | Cylinder<br>Liner                    | Refer to<br>Procedure<br>001-028<br>(/qs3/pubsy<br>s2/xml/en/pr<br>ocedures/52<br>1/521-001-<br>028.html) |    |
|                   |                                            |         | Engine<br>Compression                | Refer to<br>Procedure<br>014-012<br>(/qs3/pubsy<br>s2/xml/en/pr<br>ocedures/52<br>1/521-014-<br>012.html) | 14 |

|                |                     |         |                                |                                                                                    |    |
|----------------|---------------------|---------|--------------------------------|------------------------------------------------------------------------------------|----|
| Service Manual | L9N CM2380<br>L142B | 5659916 | Cylinder Block                 | Refer to Procedure 001-026 (/qs3/pubsys2/xml/en/procedures/1050/1050-001-026.html) | 1  |
|                |                     |         | Cylinder Block and Liner Seats | Refer to Procedure 001-027 (/qs3/pubsys2/xml/en/procedures/1050/1050-001-027.html) |    |
|                |                     |         | Cylinder Liner                 | Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/1050/1050-001-028.html) |    |
|                |                     |         | Engine Compression             | Refer to Procedure 014-012 (/qs3/pubsys2/xml/en/procedures/1050/1050-014-012.html) | 14 |

### Diagnostic Procedures

Do **not** remove the cylinder head without first performing a cylinder compression test. See corresponding Service Manual. Reference Procedure 014-012 in Section 14.

Visual inspection of a cylinder liner alone should **not** result in replacement. Visual defects **must** also correlate with a physical defect or out of specification measurement. For example, a failed visual inspection must be paired with:

- A defect that can be caught with a fingernail
- An out of specification compression test
- Cylinder liner reuse measurements out of specification

Cylinder liners **must** be clean and dry for an accurate inspection. Presence of oil, coolant, and other debris or contaminants from cylinder head removal may cause misinterpretation of cylinder liner condition. See the Clean and Inspect for Reuse section of the corresponding Service Manual. Reference Procedure 001-028 in Section 1.

Modern cylinder liner finishes may falsely appear as moderately polished or heavily polished, even when there is no wear present, depending on angle of the light source and reflections. When inspecting cylinder liners, move a focused light source around multiple locations above and in the cylinder liner to check for presence of cross hatching in all locations. It is **not** possible to inspect all areas of an installed cylinder liner with a single fixed light source. Most angles of light source produce a reflection that may give a false appearance of cylinder liner polishing across large regions of the cylinder liner. This makes photographing the cross hatching of a liner difficult. For this reason, cross hatching is **only** visible in Figures 1 through 9 across some regions of the reusable cylinder liners shown in this document.

Scratches, discoloration, and/or imperfections that are **not** able to be felt with a fingernail do **not** impact oil consumption or compression and do **not** warrant cylinder liner replacement.

## Definitions

### Moderate Polish

- A moderate polish produces a bright mirror finish in the worn area with traces of the original hone marks or an indication of an etch pattern. Moderate polishing appearance alone does **not** warrant cylinder liner replacement.

### Heavy Polish

- A heavy polish produces a bright mirror finish in the worn area with **no** traces of hone marks or an etch pattern regardless of what angle the light is. Heavy polish **must** be present around a minimum percentage of the bore circumference and minimum percentage of the piston ring travel area to warrant cylinder liner replacement. See the Clean and Inspect for Reuse section of the corresponding Service Manual. Reference Procedure 001-028 Cylinder Liner in Section 1.
- Heavy polish is normally an indication of debris ingestion also known as “dusting”. Dusting is a non-warrantable failure mode.

## Reuse Guidelines

See figures below for cylinder liner wear conditions, cause, and the corresponding action.



Figure 1

| Figure 1, New Cylinder Liner – No Surface Wear Present |                                                                                                                                                                                                                        |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance:                                            | Cross hatching clearly visible in region of proper lighting. No wear present on any part of the machined surface. Notice that cross hatching has a mirrored appearance in lower region due to lack of proper lighting. |
| Cause:                                                 | -                                                                                                                                                                                                                      |
| Action:                                                | None                                                                                                                                                                                                                   |





Figure 2

**Figure 2, Used Cylinder Liner – No Surface Wear Present**

|             |                                                                                                                                                                                                                                                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance: | Crosshatching clearly visible in region of proper lighting. Cross hatching difficult to see in regions without proper lighting. No wear present on any part of the machined surface. Notice cross hatching has a mirrored appearance in lower region due to lack of proper lighting. There is not any polishing present. |
| Cause:      | -                                                                                                                                                                                                                                                                                                                        |
| Action:     | Unconditional Reuse                                                                                                                                                                                                                                                                                                      |



Figure 3

**Figure 3, Used Cylinder Liner – Light Vertical Streaking**

|             |                                                                                                                                                                 |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance: | Narrow lines of vertical discoloration along path of piston travel. Only visible, cannot be felt with a fingernail. Crosshatching completely intact throughout. |
| Cause:      | Can occur during break-in as piston rings seat to cylinder liners.                                                                                              |
| Action:     | Unconditional Reuse                                                                                                                                             |

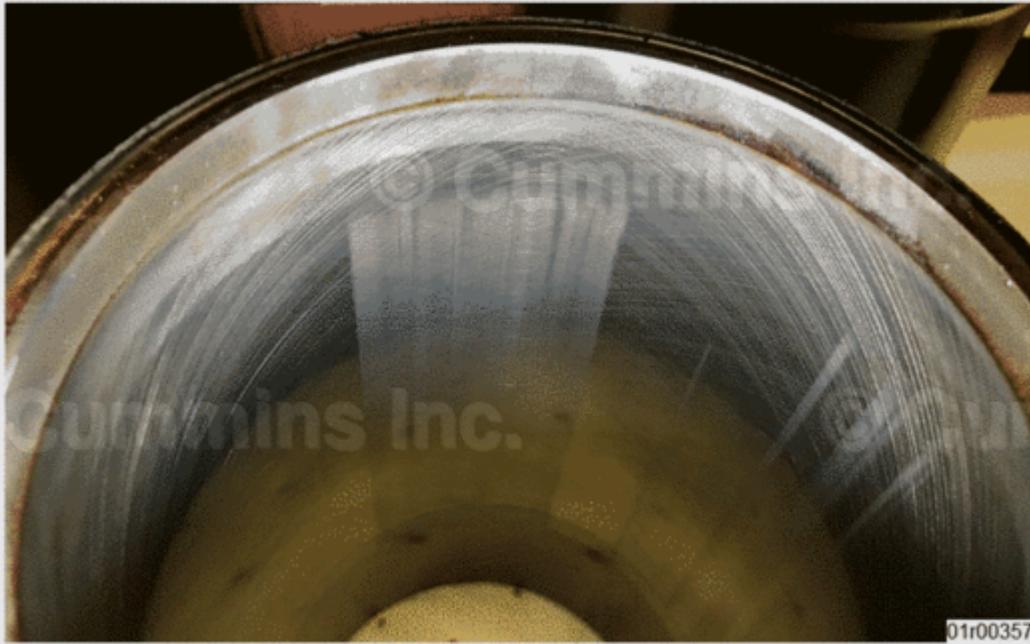


Figure 4

**Figure 4, Used Cylinder Liner – Vertical Ghosting**

|             |                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance: | Wide region of vertical discoloration along path of piston travel. Only visible, not able to be felt with a fingernail. Crosshatching completely intact throughout. |
| Cause:      | Can occur during break-in as piston rings seat to cylinder liners.                                                                                                  |
| Action:     | Unconditional Reuse                                                                                                                                                 |

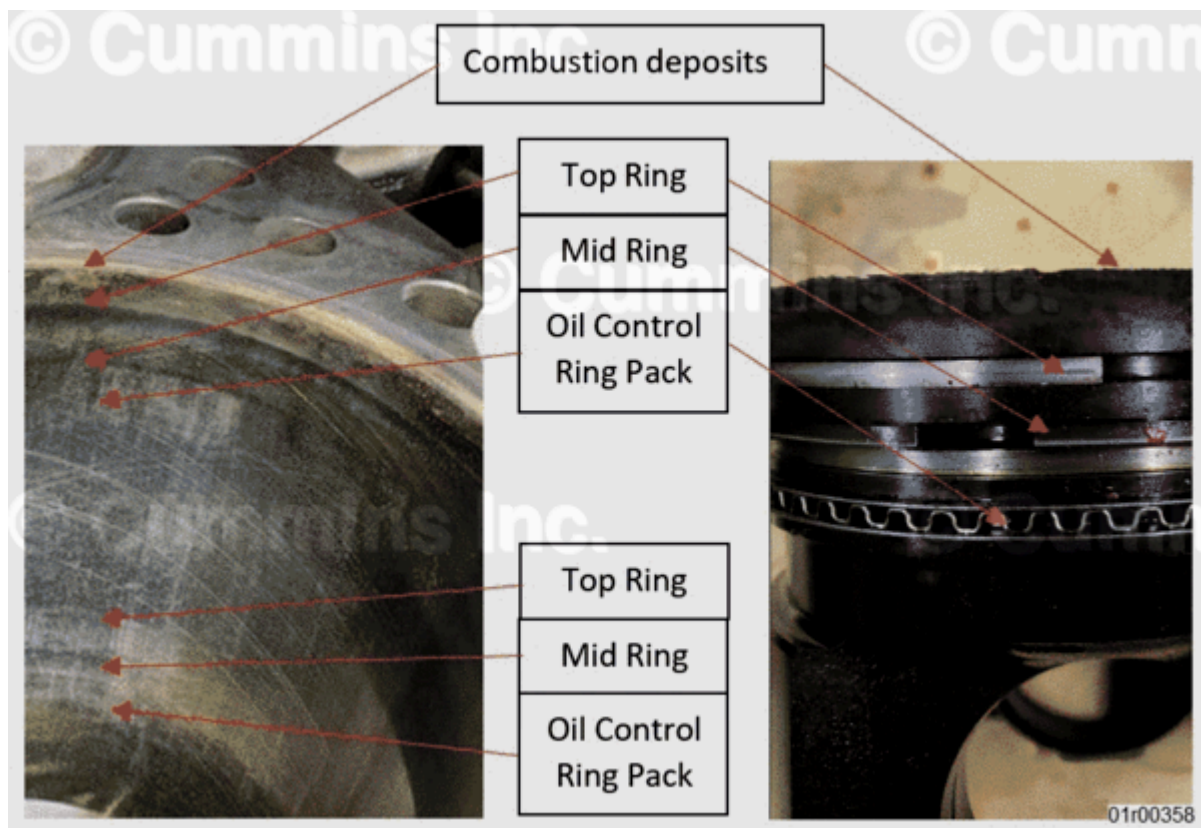


Figure 5

**Figure 5, Used Cylinder Liner – Ring Reversal Area**

|             |                                                                                                                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance: | Figure 5. Visible wear rings around circumference of cylinder liner in region of piston ring reversal, both upper and lower end of piston stroke. Not able to be felt with a fingernail. Crosshatching completely intact. |
| Cause:      | Piston rings break through oil film when changing directions at top and bottom of stroke.                                                                                                                                 |
| Action:     | Unconditional Reuse                                                                                                                                                                                                       |





Figure 6

**Figure 6, Used Cylinder Liner – Discoloration/Stain**

|             |                                                                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance: | Superficial discoloration resembling rust or hammered pattern. Not able to be felt with a fingernail. Crosshatching is difficult to see but is completely intact. |
| Cause:      | Condensation in cylinder for prolonged period. Normal for natural gas engines.                                                                                    |
| Action:     | Unconditional Reuse                                                                                                                                               |



Figure 7

**Figure 7, New or Used Cylinder Liner – Rust Before and After Running In**

|             |                                                                                                                                                                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance: | Rust colored patches. Before running in they may be felt with a fingernail, but will clean up with lubrication and run in. Not able to be felt with fingernail after run in. Cross hatching is difficult to see but is completely intact throughout discolored patch. |
| Cause:      | Condensation in cylinder for prolonged period. Normal for natural gas engines.                                                                                                                                                                                        |
| Action:     | Unconditional Reuse                                                                                                                                                                                                                                                   |



Figure 8

**Figure 8, Used Cylinder Liner – Combination of Visual Anomalies**

|             |                                                                                                                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance: | Bright spots on thrust side, staining, visible ring reversal area. No imperfections can be felt with a fingernail. Cross hatch is visible throughout with proper lighting. Notice lower region appears to have a mirrored finish due to poor lighting. |
| Cause:      | Condensation in cylinder for prolonged period. Normal for natural gas engines.                                                                                                                                                                         |
| Action:     | Unconditional Reuse                                                                                                                                                                                                                                    |



Figure 9

**Figure 9, Used Cylinder Liner – Color Variation Top to Bottom**

|             |                                                                                                                                                                                                                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Appearance: | Shown from bottom side of a removed cylinder liner for reference. Visual difference in coloring of cylinder liner from top to bottom. The darker color is not an indication of excessive wear or polishing. Cross hatching is present in all regions. Any mirrored appearance is caused by poor lighting. |
| Cause:      | Normal piston ring and cylinder liner seating.                                                                                                                                                                                                                                                            |
| Action:     | Unconditional Reuse                                                                                                                                                                                                                                                                                       |

## Document History

| Date      | Details        |
|-----------|----------------|
| 2021-4-14 | Module Created |



